Volume by Slicing: Fubini, Cavalieri, and Coarea

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Every triple integral starts with a region. Choosing the order of integration means choosing how that region will be decomposed into fibers: vertical columns, horizontal slices, or curved level sets. Fubini's theorem guarantees that legitimate decompositions give the same integral; geometry determines which decomposition is useful. The connection with probability and the coarea formula arises because every slicing operation pushes volume onto a lower-dimensional set.[1]

§1.1 A region is its indicator function

Let $\Omega \subset \mathbb{R}^3$ be a measurable bounded solid. Its volume is

$$|\Omega| = \iiint_{\mathbb{R}^3} \mathbf{1}_\Omega(x, y, z)\, dx\, dy\, dz.$$

For fixed (x, y), the vertical fiber is

$$\Omega_{x,y} = \{z : (x, y, z) \in \Omega\}.$$

If almost every such fiber is the interval $[z_-(x, y), z_+(x, y)]$, Tonelli's theorem gives

$$|\Omega| = \iint_D (z_+(x, y) - z_-(x, y))\, dx\, dy,$$

where D is the projection of the solid onto the xy-plane.

The height formula is therefore an integral of a one-dimensional fiber length. In the opposite direction, define the horizontal cross-sectional area

$$A(z) = \iint_{\mathbb{R}^2} \mathbf{1}_\Omega(x, y, z)\, dx\, dy.$$

The very same theorem gives

$$\boxed{|\Omega| = \int_{\mathbb{R}} A(z)\, dz.}$$

These are two different coordinate descriptions of the same nonnegative indicator function; no formal permutation of the integration symbols is needed.

§1.2 One paraboloid, two calculations

Consider

$$\Omega = \{(x, y, z) : x^2 + y^2 \le 4, \quad 0 \le z \le 4 - x^2 - y^2\}.$$

Vertical fibers have height $4 - r^2$. Using polar coordinates in the projection plane,

$$|\Omega| = \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr\, d\theta = 8\pi.$$

[1] For Fubini–Tonelli and integration along level sets, see Evans and Gariepy, *Measure Theory and Fine Properties of Functions*, and MIT 18.02SC's multiple-integral material: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/.

Now fix z. The inequality $z \le 4 - x^2 - y^2$ describes a disk of radius $\sqrt{4-z}$, with area $A(z) = \pi(4-z)$ for $0 \le z \le 4$. Thus

$$|\Omega| = \int_0^4 \pi(4-z)\,dz = 8\pi.$$

The first calculation integrates vertical column heights; the second integrates horizontal disk areas. Their equality is Cavalieri's principle made quantitative.

§1.3 Layer cake turns height into level sets

For a nonnegative measurable function $h : \mathbb{R}^n \to [0,\infty]$, the identity

$$h(x) = \int_0^\infty \mathbf{1}_{\{h(x)>t\}}\,dt$$

holds pointwise, with extended values allowed. Integrating and applying Tonelli gives the *layer-cake formula*

$$\boxed{\int_{\mathbb{R}^n} h(x)\,dx = \int_0^\infty |\{x : h(x) > t\}|\,dt.}$$

This is the general principle behind the paraboloid example: the superlevel set of $h(x,y) = \max(4 - x^2 - y^2, 0)$ at height $t \in (0,4)$ is a disk of area $\pi(4-t)$. The integral of those areas again equals 8π.

The formula is also useful when the distribution of heights is known but no explicit parametrization is convenient. In probability, if $X \ge 0$ is a random variable,

$$\mathbb{E}[X] = \int_0^\infty \Pr(X > t)\,dt,$$

the same identity with probability measure replacing area. Cross-sectional area and survival probability are two instances of integrating fiber masses.

§1.4 Choosing Effective Slices

The intersection of two perpendicular unit cylinders is the solid

$$\Omega = \{(x,y,z) : x^2 + z^2 \le 1, \quad y^2 + z^2 \le 1\}.$$

Integrating first in x and y gives an awkward description if written without examining the intersection. But for fixed $z \in [-1,1]$, its cross-section is simply a square:

$$|x| \le \sqrt{1-z^2}, \qquad |y| \le \sqrt{1-z^2}.$$

The side length is $2\sqrt{1-z^2}$, so its area is $4(1-z^2)$. Therefore

$$\boxed{|\Omega| = \int_{-1}^1 4(1-z^2)\,dz = \frac{16}{3}.}$$

This classical Steinmetz solid illustrates why a carefully chosen slicing direction can replace a difficult curved region by elementary cross-sections.

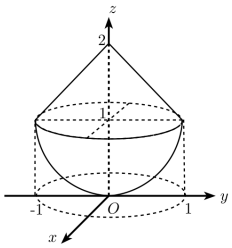


Figure 1.1: A composite solid motivates choosing its cross-sections before committing to an order of integration.

§1.5 Projection changes can force a split of the domain

A region may be simple when projected onto one coordinate plane but require several pieces in another projection. For example, the tetrahedron

$$T = \{(x, y, z) : x, y, z \geq 0, \quad x + y + z \leq 1\}$$

has vertical fibers $0 \leq z \leq 1 - x - y$ above the triangle $x, y \geq 0$, $x + y \leq 1$. Thus

$$|T| = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \frac{1}{6}.$$

Its horizontal slice at height z is instead a right triangle with legs $1-z$, of area $(1-z)^2/2$, so

$$|T| = \int_0^1 \frac{(1 - z)^2}{2}\, dz = \frac{1}{6}.$$

A more complicated region may have a cross-section whose boundary changes from a graph to a circle or whose number of connected components changes. At such transition levels, the formula for the fiber area changes, and the outer integral must be split accordingly. The need to split arises from actual geometry, not from the syntax of an iterated integral.

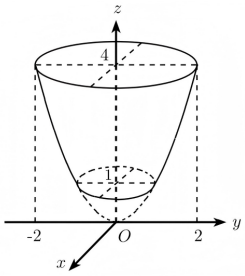


Figure 1.2: A truncated rotational solid may be more naturally described by radial or horizontal slices than by vertical columns.

§1.6 Coarea is slicing along nonlinear level sets

Horizontal slicing uses the function $g(x, y, z) = z$, whose level sets are planes and whose gradient has norm one. The coarea theorem generalizes this idea: for a sufficiently regular $g : \mathbb{R}^n \to \mathbb{R}$ and suitable nonnegative h,

$$\int_{\mathbb{R}^n} h(x) |\nabla g(x)|\, dx = \int_{\mathbb{R}} \left(\int_{g^{-1}(t)} h(x)\, d\mathcal{H}^{n-1}(x) \right) dt.$$

Here $\mathcal{H}^{n-1}$ is the natural surface measure on the level set. The factor $|\nabla g|$ accounts for how densely the levels are packed.

For $g(x, y) = \sqrt{x^2 + y^2} = r$ in the plane, $|\nabla g| = 1$ away from the origin. The level sets are circles, with arc-length element $r\, d\theta$. Coarea therefore gives polar integration:

$$\int_{\mathbb{R}^2} h(x, y)\, dx\, dy = \int_0^{\infty} \int_0^{2\pi} h(r\cos\theta, r\sin\theta)\, r\, d\theta\, dr.$$

The familiar polar Jacobian r is the length scaling of a circular level set, rather than an arbitrary factor to memorize.

Vertical integration, Cavalieri's horizontal slices, layer cake and coarea all express the same operation: a measure is disassembled along fibers of a map. For z-projection the fibers are plane sections, for a height function they are superlevel sets, and for a radius map they are circles. The art of changing integration order is to choose the map whose fibers have the simplest geometry.